Assignment Spring 2025 Report

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Subject: COMP7024 - Programming for Data Science

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Project Assumptions:

> The analysis will use the provided CSV files by the assignment. For a comprehensive analysis, the PG1 and PG2 files will be combined into two one data frames for each type of dataset (`patients`. `encounters`, `conditions`). All data files will be included in the repository. The analysis is performed using R within an R Markdown document, which allows for reproducible research by combining code, results, and narrative. As per the project context, all patient ages will be calculated relative to the date **October 13, 2025**. The code will use the column names as they appear in the CSV files (e.g., `Id`, `BIRTHDATE`, `COUNTY`, `PATIENT`, `DESCRIPTION`).

The code in this report have been ensured to run without errors. The codes are well-commented and separated into sections for clarity. All warning and formatting issues have been addressed to ensure a clean and professional presentation.

We first needs to loads all packages required for the assignment, including data manipulation, date calculations, and plot arrangement, and other utilities.

```r
knitr::opts_chunk$set(echo = TRUE)

library(tidyverse)
library(lubridate)
library(patchwork)
library(kableExtra)
```

Task 1

The goal is to analyze and visualize the distribution of patients based on two criteria: 1. The geographic distribution of patients diagnosed with COVID-19 (confirmed or suspected) across different counties. 2. The demographic distribution of the entire patient population across four age groups: 0-18, 19-35, 36-50, and 51+.

Data preparation We then handles the loading and preparation of the data. The script explicitly loads the four CSV files and combines them using `rbind()` so that the two patient files are merged as one, and the two condition files are also merged as one.

```r
patients_pg1 <- read.csv("patientsPG1.csv")
patients_pg2 <- read.csv("patientsPG2.csv")

patients <- rbind(patients_pg1, patients_pg2)

conditions_pg1 <- read.csv("conditionsPG1.csv")
conditions_pg2 <- read.csv("conditionsPG2.csv")

conditions <- rbind(conditions_pg1, conditions_pg2)
```

Now, we perform the analysis using the combined data frames.

The script first isolates the patient IDs associated with a "COVID" diagnosis and then uses those IDs to filter the main patient list, allowing for an accurate count by county. The script also uses `mutate()` to add new columns. Ages are calculated precisely using `lubridate`, and the `cut()` function is used to efficiently categorize all patients into the defined age bins.

```r
# --- Distribution of COVID Patients by County ---
covid_patient_ids <- conditions %>%
  filter(grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
  distinct(PATIENT) %>%
  pull(PATIENT)

covid_by_county <- patients %>%
  filter(Id %in% covid_patient_ids) %>%
  count(COUNTY, name = "patient_count") %>%
  arrange(desc(patient_count))


# --- Distribution of Patients by Age Group ---
today <- as.Date("2025-10-13")

patients_by_age_group <- patients %>%
  mutate(
    BirthDate = as.Date(BIRTHDATE),
    Age = floor(time_length(interval(BirthDate, today), "years")),
    `Age Group` = cut(Age,
                      breaks = c(-1, 18, 35, 50, Inf),
                      labels = c("0-18", "19-35", "36-50", "51+"),
                      right = TRUE)
  ) %>%
  count(`Age Group`, name = "patient_count")
```

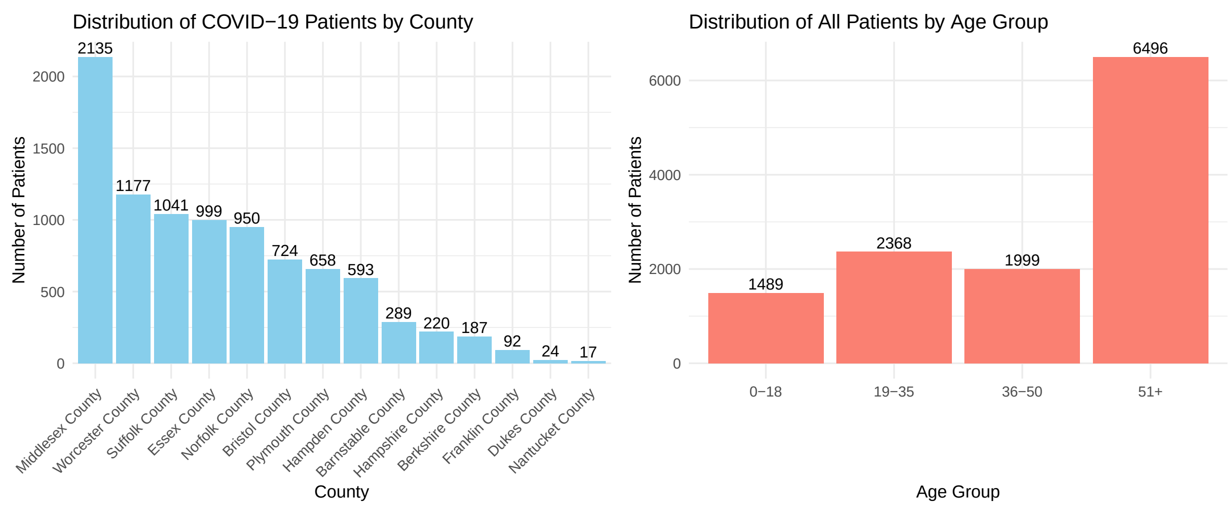
Visualization After preparing the data, we create two bar plots using `ggplot2`. The first plot visualizes the distribution of COVID-19 patients by county, while the second plot shows the distribution of all patients by age group. The plots are then arranged side-by-side using the `patchwork` library for better comparison.

Both distributions were visualized using bar plots instead of histograms, as bar plots are more appropriate for categorical data like counties and age groups.

```r
# Plot 1: COVID Patients by County
plot1 <- ggplot(covid_by_county,
                aes(x = reorder(COUNTY, -patient_count), y = patient_count)) +
  geom_col(fill = "skyblue") +
  geom_text(aes(label = patient_count), vjust = -0.3) +
  labs(title = "Distribution of COVID-19 Patients by County",
       x = "County",
       y = "Number of Patients") +
  theme_minimal(base_size = 12) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# Plot 2: All Patients by Age Group
plot2 <- ggplot(patients_by_age_group,
                aes(x = `Age Group`, y = patient_count)) +
  geom_col(fill = "salmon") +
  geom_text(aes(label = patient_count), vjust = -0.3) +
  labs(title = "Distribution of All Patients by Age Group",
       x = "Age Group",
       y = "Number of Patients") +
  theme_minimal(base_size = 12)
```

```r
# Arrange the two plots side-by-side using the patchwork library.
plot1 + plot2
```



The bar chart on the left shows the absolute number of patients with a COVID-19 diagnosis across the different counties in Massachusetts. There is a significant geographic concentration of COVID-19 cases. Middlesex County stands out as the area with the highest number of patients, totaling 2,135. This figure is nearly double that of the next highest county, Worcester.

The top eight counties collectively account for the vast majority of the COVID-19 cases in this dataset. This suggests that these are major hotspots. This information indicates that medical resources and public health interventions may need to be concentrated in these areas to effectively manage and contain the spread of the virus.

The bar chart on the right provides a demographic snapshot of the entire patient population. The patient population is heavily skewed towards older adults. The 51+ age group is by far the largest demographic, with 6,496 patients. This is more than the other three age groups combined. This distribution strongly suggests that older adults are more likely to seek medical care or have more health issues requiring attention. There is a surprising spike when people reach the age of 51 compared to the 36-50 age group. This could be due to a variety of factors, including the onset of age-related health conditions that become more prevalent after 50 or that people are more proactive about their health as they age. This information means that more pressure may be placed on healthcare services that cater to older adults, including chronic disease management and preventive care.

Task 2

1. Identify the top 5 most common conditions for all COVID-19 patients, excluding conditions that are directly related to the COVID-19 diagnosis itself.
2. Investigate if these common co-occurring conditions differ by gender.
3. Present the top 10 conditions for male and female patients separately in a formatted table for comparison and provide an interpretation of the findings.

Data preparation For task 2, we reuse the `covid_patient_ids` vector created in Task 1 to filter the `conditions` data frame. This ensures that we are only analyzing conditions for patients who have had a COVID-19 diagnosis.

The code block below uses a `dplyr` pipe to chain operations. It `filter()`s for our COVID patient cohort, then uses `filter()` with a `!` (negation) to *remove* all COVID-related diagnoses. Finally, `count()` tallies the remaining conditions, and `top_n(5)` selects the five most frequent ones. A crucial step here is the `inner_join()`, which merges the two data frames, allowing us to analyze conditions and demographics together. The logic is then repeated twice, adding a `filter()` for `GENDER == 'M'` and `GENDER == 'F'` respectively to create two separate top-10 lists.

```
# --- Find Top 5 Most Common Conditions (Overall) ---
top_5_conditions_overall <- conditions %>%
  filter(PATIENT %in% covid_patient_ids) %>%
  filter(!grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
  count(DESCRIPTION, sort = TRUE, name = "count") %>%
  top_n(5, count)

print(top_5_conditions_overall)
```

```
##                                DESCRIPTION count
## 1                          Fever (finding)  8083
## 2                          Cough (finding)  6202
## 3                  Loss of taste (finding)  4711
## 4                        Fatigue (finding)  3516
## 5 Body mass index 30+ - obesity (finding)  3490
```

```
# --- Analyze Top 10 Conditions by Gender ---

conditions_with_gender <- conditions %>%
  inner_join(patients, by = c("PATIENT" = "Id"))

top_10_male <- conditions_with_gender %>%
  filter(PATIENT %in% covid_patient_ids, GENDER == 'M') %>%
  filter(!grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
  count(DESCRIPTION, sort = TRUE, name = "count") %>%
  head(10)

top_10_female <- conditions_with_gender %>%
  filter(PATIENT %in% covid_patient_ids, GENDER == 'F') %>%
  filter(!grepl("COVID", DESCRIPTION, ignore.case = TRUE)) %>%
  count(DESCRIPTION, sort = TRUE, name = "count") %>%
  head(10)
```

The code block above prints the overall top 5 health issues experienced by COVID-19 patients in this dataset. Fever is the most common condition with a count of 8083, followed by Cough, Loss of taste, Fatigue, and obesity with a body mass index over 30. Identifying these conditions is crucial for patient management, as it points to which other conditions doctors should monitor closely in a COVID-19 patient.

Visualization The next code block produces a side-by-side comparison of the top 10 conditions for male and female patients.

The data are first prepared by renaming columns for clarity. The `cbind()` function is then used to combine the two data frames into one for side-by-side comparison. Finally, `kbl()` and `kable_styling()` are used to format the table with a caption and styling options.

Table 1: Top 10 Most Common Non-COVID Conditions in COVID-19 Patients by Gender

Male Patients		Female Patients	
Male Condition	M_Count	Female Condition	F_Count
Fever (finding)	3833	Fever (finding)	4250
Cough (finding)	2936	Cough (finding)	3266
Loss of taste (finding)	2271	Loss of taste (finding)	2440
Fatigue (finding)	1695	Body mass index 30+ - obesity (finding)	1918
Body mass index 30+ - obesity (finding)	1572	Fatigue (finding)	1821
Sputum finding (finding)	1406	Miscarriage in first trimester	1580
Anemia (disorder)	1387	Sputum finding (finding)	1564
Prediabetes	1302	Prediabetes	1377
Hypertension	1084	Hypertension	1183
Hypoxemia (disorder)	865	Normal pregnancy	1063

```r
# --- Present the Gender Comparison in a Table ---

male_table_data <-
  top_10_male %>% rename(`Male Condition` = DESCRIPTION, `M_Count` = count)
female_table_data <-
  top_10_female %>% rename(`Female Condition` = DESCRIPTION, `F_Count` = count)

comparison_table <- cbind(male_table_data, female_table_data)

comparison_table %>%
  kbl(caption = "Top 10 Most Common Non-COVID
   Conditions in COVID-19 Patients by Gender") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                full_width = FALSE) %>%
  add_header_above(c("Male Patients" = 2, "Female Patients" = 2))
```

According to the table above, eight of the top 10 conditions are shared between male and female patients. Classic viral symptoms such as Fever, Cough, Loss of taste, and Fatigue rank in the top 5 for both genders. This is consistent with the well-documented presentation of COVID-19 and confirms the symptomatic overlap between men and women.

The presence of Obesity (BMI 30+), Prediabetes, and Hypertension on both lists is highly significant. These conditions are widely recognized as major risk factors that increase the likelihood of severe complications from COVID-19. Their high frequency underscores the role of metabolic and cardiovascular health in disease outcomes for all patients.

For male, the unique conditions points towards severe respiratory issues. The appearance of Hypoxemia, or low blood oxygen levels, is a direct indicator of this concern. While anemia can be associated with chronic inflammation and more sever systemic illness, it may reflect that men were experiencing an inflammatory response to COVID-19.

For female, the unique conditions points towards pregnancy-related issues. The high pressence of miscarriage is alraming. This finding emphasizes the need for special care and monitoring of pregnant patients with COVID-19, as they may face unique risks and complications. However, there is also the presence of normal pregnancy, which suggests that not all pregnant patiens with COVID-19 experience adverse outcomes, and that the dataset highlight that pregnant women are more likely to contracts COVID-19 compared to other demographics.